SEGA-NET: LLM-Guided Semantic-Enhanced GAN Augmentation Network for Low-Resolution Image Classification

Mohammad Shahedur Rahman



Mohammad Tahmid Bari



Md Delwar Hossain



Roya Choupani

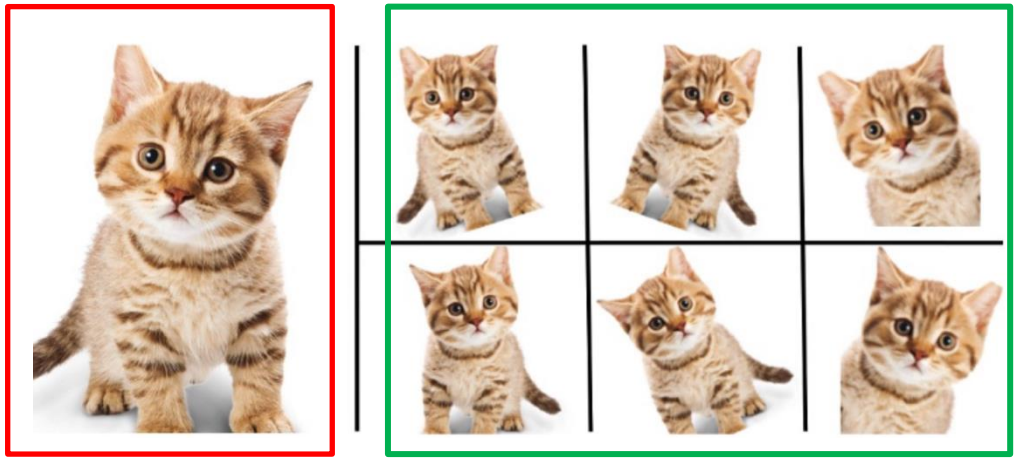


Erdogan Dogdu



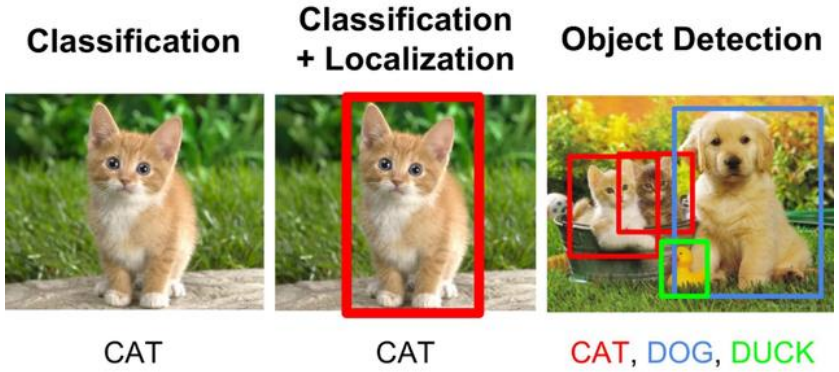
Data Augmentation

- Data augmentation refers to
  - a technique that artificially expand the training data
  - applying label-preserving transformation to improve generalization and robustness.
    - Geometric: flip, crop, rotation, scaling
    - Photometric: color jitter, noise, blur.
    - Mixing-based: Mixup, CutMix, SaliencyMix.
    - Generative: GAN-based, diffusion-based.



Why Data Augmentation Matters for Low-Resolution Vision?

- Motivation:
  - Limited semantic capacity and overfitting: it lacks *semantic context*, causing models to overfit to *spurious textures* and *local patterns*.



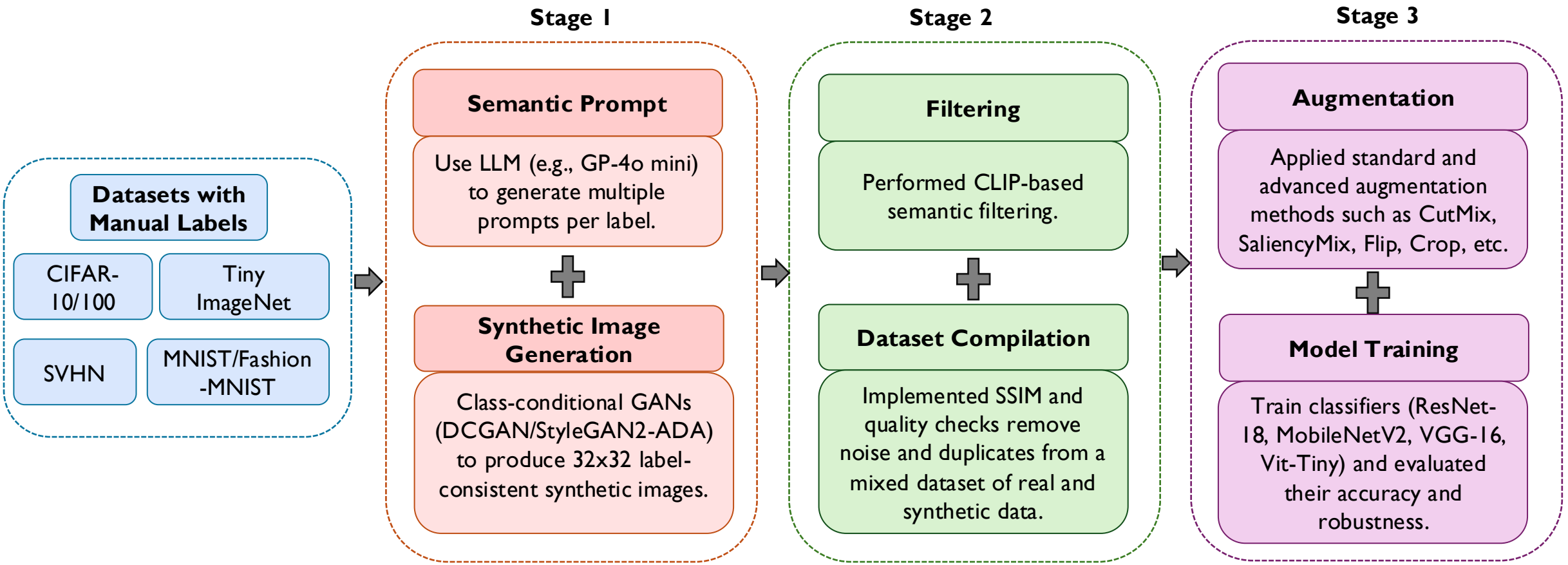
  - Limited effectiveness of standard augmentation: They offer *pixel-level variations only*, failing to introduce *semantic diversity*.
  - Limited robustness and generalization: Models trained on low-resolution data show *degraded robustness* under *distribution shifts* and *adversarial attacks*.

- *An effective study on data augmentation is required for the low-resolution vision.*

Our Contribution: SEGA-NET

- SEGA-Net: a semantic-aware three-stage augmentation framework for efficient and robust low resolution image classification.
  - Semantic-guided prompt generation
    - *First framework* to *bridge* LLM semantics with class-conditional GANs.
  - Security-aware semantic filtering
    - Apply *CLIP-based alignment, SSIM de-duplication* and *quality-check* to retain label-consistent, high-fidelity synthetic samples.
  - Robust model training
    - Merge filtered synthetic data with real data and apply *pixel-level augmentation*.

Overview of SEGA-NET



SEGA-NET: Semantic-Guided Prompt Generation

- LLM-based expansion
  - Expand each class label into *multiple semantically rich textual prompts* using an LLM (GPT-4o-mini).
- Text-to-latent semantic bridging
  - Map CLIP-embedded prompts into the GAN latent space via a *lightweight text-to-latent projector*.
- Efficient low-resolution synthesis
  - Steer a *frozen class-conditional GAN* to generate *label-consistent 32x32 synthetic images* with low computational cost.

Algorithm 1 (Stage-1: Training the prompt→latent projector P_θ (with G frozen)

Require: Frozen class-conditional GAN G (pretrained per dataset); CLIP encoders (ϕ_v, ϕ_t); class prompts $\{\mathcal{P}_y\}$; clusters $\{C_{y,j}\}$ (from cosine similarity, m=3 fixed); noise scale σ; temperature τ; regularization weight λ; learning rate η; total steps T

Ensure: Projector $P_\theta : \mathbb{R}^{d_{\text{text}}} \to \mathbb{R}^{d_z}$

Initialize MLP projector $P_\theta : d_{\text{text}} \to 1024 \to d_z$ with GELU and LayerNorm

for $t = 1$ to T **do**

  Sample a class y and a cluster $C_{y,j}$; form pair $(\bar{e}_{y,j}, p^+)$ with $p^+ \in C_{y,j}$

  Compute latent vector: $z \leftarrow P_\theta(\bar{e}_{y,j}) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I)$

  Generate synthetic image: $x \leftarrow G(z, y)$ ▷ Forward pass only; G frozen

  Compute CLIP contrastive loss (InfoNCE): $\mathcal{L}_{\text{CLIP}}$

  Compute regularization term: $\mathcal{L}_{\text{reg}} = \lambda \| P_\theta(\bar{e}_{y,j}) \|_2^2$

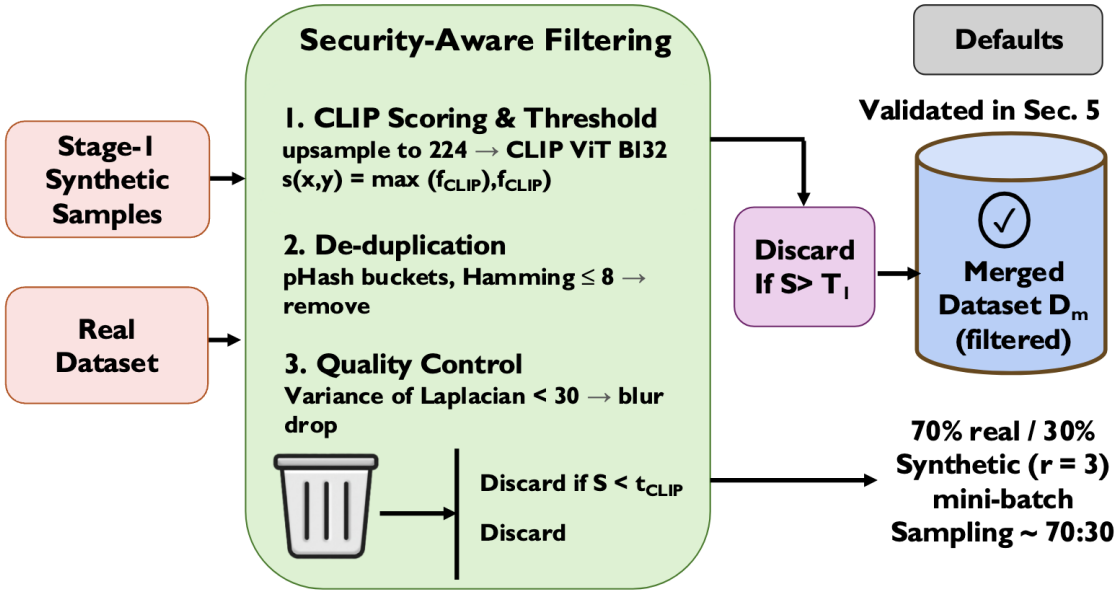
  Update projector parameters:

$$\theta \leftarrow \theta - \eta \nabla_\theta \left(\mathcal{L}_{\text{CLIP}} + \mathcal{L}_{\text{reg}} \right)$$

end for

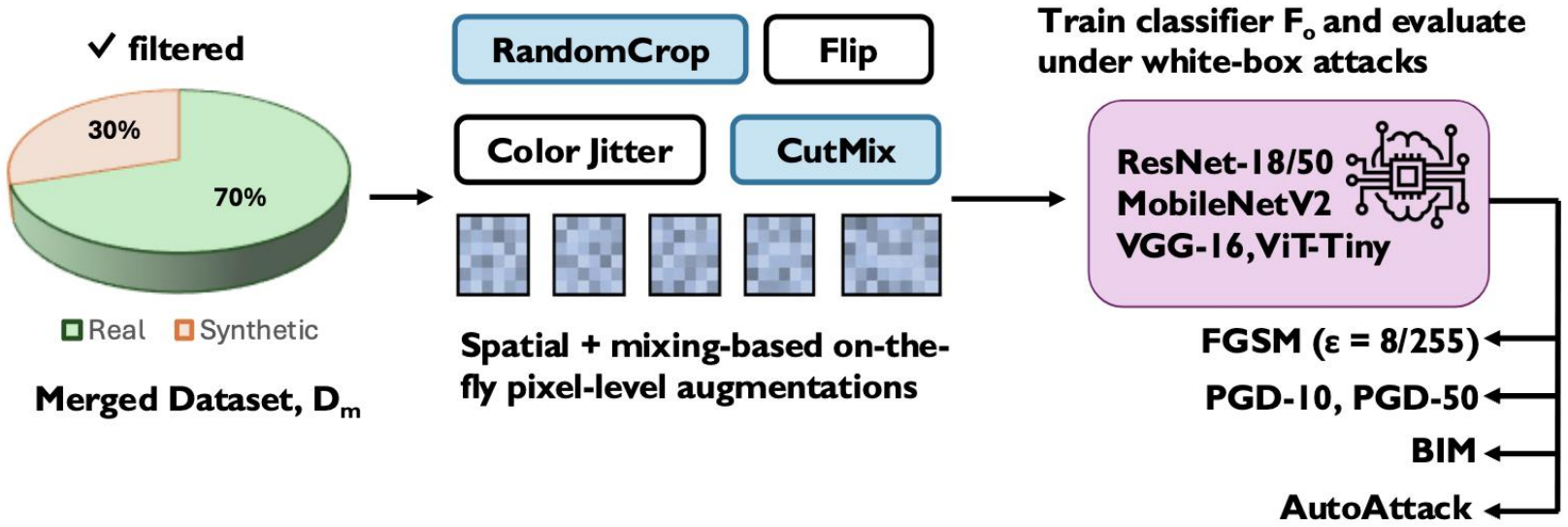
SEGA-NET: Security-Aware Semantic Filtering

- Sematic alignment validation
  - Use *CLIP-based similarity scoring* to ensure generated images align with class semantics.
- Redundancy and quality control
  - Remove near-duplicates using *pHash + SSIM* and discard low-quality samples via *blur detection.*
- Clean synthetic dataset construction
  - Retain only *high-fidelity, label-consistent synthetic samples* for downstream training.



SEGA-NET: Robust Training with Augmentation

- Real-synthetic data integration
  - Merge filtered synthetic samples with real data using a *controlled ratio* to preserve class balance.
- Augmented and robust learning
  - Train classifier with *pixel-based* and *mixing-based* augmentations to improve *accuracy, robustness,* and *generalization.*



SEGA-NET: Experimental Setup

- Dataset and Models
  - CIFAR 10/100, Tiny-Imagenet, Fashion-MNIST (32x32) with ResNet, MobileNetv2, VGG-16 and ViT-Tiny.

Dataset	Cls	Train	Test	N_{syn}
CIFAR-10	10	50,000	10,000	21,428
CIFAR-100	10	50,000	10,000	21,428
Tiny-ImageNet	200	100,000	10k	42,857
SVHN	10	73,257	26,023	31,498
Fashion-MNIST	10	60,000	10,000	25,714

- Training and Augmentation
  - 70:30 real-synthetic with pixel-level augmentations, trained for 300 epochs under identical budgets.
- Evaluation Matrices
  - Measured clean-accuracy, robustness to adversarial attacks (FGSM, PGD, BIM, AutoAttack) and generalization under distribution shifts.

SEGA-NET: Experimental Findings

- We comprehensively evaluated SEGA-NET across six complementary dimensions:
  - Clean accuracy evaluation
  - Generalization under domain shifts
  - Adversarial robustness evaluation
  - Semantic alignment evaluation
  - Scalability and efficiency analysis
  - Sensitivity and ablation studies

SEGA-NET: Clean Accuracy Evaluation

- Verified that semantic-augmentation improves standard classification accuracy under identical training budgets.

- SEGA-Net improves by +8-10% over standard augmentation across all dataset.

Method	CIFAR10	CIFAR 100	Tiny ImageNet	SVHN	Fashion MNIST
No Aug	68.5±0.6	48.5±0.5	35.2±0.5	83.9±0.4	90.6±0.3
Std. Aug	73.2±0.5	50.4±0.6	36.9±0.5	88.1±0.3	93.2±0.3
GAN-only	74.0±0.4	51.8±0.5	37.2±0.6	89.2±0.4	94.0±0.2
Stable Diff.	74.5±0.5	52.1±0.5	37.8±0.6	90.1±0.3	94.7±0.2
SEGA-NET	**78.3±0.4**	**53.7±0.5**	**39.2±0.6**	**92.0±0.3**	**96.5±0.2**

SEGA-NET: Generalization Under Domain Shifts

- Tested robustness under corruptions and cross-domain-shifts without retraining.

- SEGA-Net improves accuracy by +5.9% over standard augmentation and +10.3% over No augmentation on CIFAR-10-C dataset.

Method	CIFAR 10-C	CIFAR 100-C	Tiny IN (C)	SVHN MNIST	F-MNIST
No Aug	54.2±0.5	32.1±0.6	21.4±0.4	65.3±0.6	71.2±0.5
Std. Aug	58.7±0.6	35.4±0.5	24.8±0.5	72.0±0.5	76.5±0.4
SEGA-NET	**64.5±0.4**	**39.6±0.5**	**28.2±0.5**	**78.1±0.4**	**82.3±0.3**

SEGA-NET: Adversarial Robustness Evaluation

- Evaluated resistance to white-box adversarial attacks without adversarial training.

- SEGA-Net improves robustness by +7-12% across FGSM, PGD, BIM, AutoAttack.

Method	FGSM	PGD-10	PGD-50	BIM	AutoAttack
No Aug	43.2±0.6	29.4±0.5	25.1±0.5	31.0±0.6	23.5±0.5
Std. Aug	48.7±0.5	33.6±0.6	29.8±0.5	36.2±0.5	27.4±0.4
SEGA-NET	**55.3±0.4**	**41.2±0.5**	**36.5±0.5**	**43.8±0.4**	**34.1±0.4**

SEGA-NET: Semantic Alignment Evaluation

- Measured whether filtering improves semantic consistency of synthetic samples.

- CLIP filtering increases median similarity by +0.10-0.15 across datasets.

Method	CIFAR10	CIFAR 100	Tiny IN	SVHN	F-MNIST
Pre-filter	0.65 / 0.18	0.55 / 0.22	0.50 / 0.25	0.65 / 0.19	0.60 / 0.21
Post-filter	**0.75 / 0.10**	**0.70 / 0.12**	**0.65 / 0.14**	**0.80 / 0.08**	**0.75 / 0.11**

SEGA-NET: Scalability and Efficiency Analysis

- Compared generation quality vs computational cost with diffusion and GAN baselines.

- SEGA-Net is +6.7x more compute efficient than diffusion-based augmentation.

Variant	sec/ img	GPU (h)	Mem (GB)	Params (M)	FID	Quality (1-5)
SEGA-NET (full)	0.45	12	16	50.2	12.5	4.3
SEGA-NET (no GAN)	0.42	11.5	15	50.2	14	4
SEGA-NET (no LLM)	0.4	11	15	50.2	13.8	4.1
Stable Diffusion	3.2	80	24	1100	8.2	4.8
Diffusion (no filter)	2.8	75	22	1100	9	4.7

SEGA-NET: Sensitivity and Ablation Studies

- Identified the contribution of LLM prompts, GAN synthesis, and filtering.

- Removing any component reduces accuracy by +3.4% average accuracy.

Method	CIFAR 10	CIFAR 100	Tiny ImageNet	SVHN	Fashion MNIST	Avg.
GAN-only	70.1	45.3	32	89.5	85	64.4
LLM-only	72	46.8	33.5	90.3	86.2	65.8
GAN+LLM	73.5	48.2	35.1	91.2	87.4	67.1
GAN+Filter	71.8	47	34	90	86.8	65.9
LLM+Filter	74.2	49	36.2	92.1	88	67.9
SEGA-NET	**78.3**	**53.7**	**39.2**	**94.5**	**90.6**	**71.3**

Key Takeaways

- *SEGA-Net*, the first semantic-aware GAN augmentation framework integrating LLM-guided prompts for low-resolution image classification.
- A principled approach to semantic augmentation using projector-steered GANs with quality-controlled filtering.
- Deep understanding of how semantic augmentation in low-resolution image classification improves generalization, robustness, and efficiency.

Thank You!

- Webpage: https://mdshahedrahman.github.io/

